

Configuration Management with Ansible/Chef/Puppet/Terraform (15 points):

1. **Choose one configuration management tool (Ansible, Chef, or Puppet, terraform). Write a script/playbook to automate the deployment of a web server on a virtual machine.**

→ YAML file for this task is in path: `tek_challenge/task-3/ansible-playbook/`

Below is the snapshot of ansible output where the simple application has been deployed over a vm using ansible-playbook:

```
$ ansible-playbook -i hosts.ini webserver_deployement.yml --ask-become-pass
BECOME password:

PLAY [Deploy Nginx web server] *****

TASK [Gathering Facts] *****
ok: [webserver01]

TASK [Install Nginx on Debian/Ubuntu] *****
changed: [webserver01]

TASK [Install Nginx on RHEL/CentOS/Amazon] *****
skipping: [webserver01]

TASK [Ensure Nginx enabled and running] *****
ok: [webserver01]

TASK [Deploy index.html (Debian/Ubuntu default path)] *****
changed: [webserver01]

TASK [Deploy index.html (RHEL/CentOS/Amazon default path)] *****
skipping: [webserver01]

PLAY RECAP *****
webserver01          : ok=4    changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

CI/CD Pipeline Configuration (25 points):

1. **Create a Jenkins pipeline configuration (Jenkinsfile) that includes stages for building, testing, and deploying a sample application to Azure. Ensure that the pipeline includes security scanning as a step.**

Jenkins and other related file are in the mentioned path: `tek_challenge/task-3/Jenkins/`

Below is the snapshot for the output of the jenkins CICD pipeline

✓ #9 (Aug 19, 2025, 4:52:31 PM)

 Add description

Keep this build forever



Started by user [admin](#)

Started 4 min 54 sec ago

Took [28 sec](#)



This run spent:

- 17 ms waiting;
- 28 sec build duration;
- 28 sec total from scheduled to completion.



Revision: 897046022d10343e8085f82ec8a2de98858c8188

Repository: <https://github.com/shahab/tek-challenge.git>

- refs/remotes/origin/main



Changes

1. tek-challenge jenkins cid ([details](#) / [githubweb](#))